Replication Study - How APIs Create Growth by Inverting the Firm

Benzell, Hersh & Van Alstyne (2024)

What Does “Inverting the Firm” Mean?

Traditional Theory of the Firm

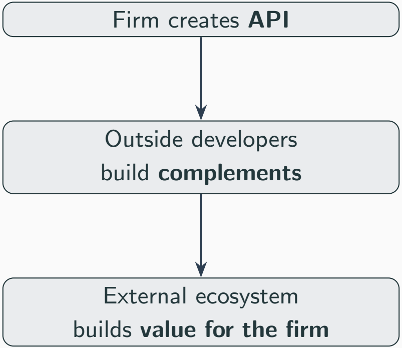
Core Idea
A firm grows by **bringing activities inside** the firm.

Examples:

- Hiring employees
- Building products internally

Inversion = Pushing Innovation Outward

Inversion
Instead of moving activities *inward*, the firm pushes innovation **outward**.



Why Would This Create Growth?

Three Mechanisms (Paper's Framework)

The paper proposes **three complementary mechanisms**:

1. More Complement Innovation
2. Lower Innovation Costs
3. Network Effects

① Mechanism 1 — More Complement Innovation

- Outside developers create products the firm *never had to build*.

$\Rightarrow$ More complements $\to$ more user value.

Classic Indirect Network Effect
A larger ecosystem of complements makes the core platform more attractive to users, which in turn attracts more developers.

Mechanism 1 — More Complement Innovation: Razorpay

Razorpay API (Core Platform)

↓

Outside Developers: Indian Startups, D2C Brands

↓

Products They Built: CRED, Ola, Unacademy, Dunzo

Key Point: Razorpay didn't build CRED. CRED built ON Razorpay.

② Mechanism 2 — Lower Innovation Costs

- External parties **bear part of the innovation cost**.
- The firm gets the upside *without* full internal investment.

Key Implication

Open innovation via APIs shifts R&D expenditure to the ecosystem, improving the firm's return on investment.

Mechanism 2 — Lower Innovation Costs: Example

Example: Google Maps API in India

Who	What They Pay For
Google	Provides base map and API infrastructure
Ola / Zomato / Swiggy	Build tracking, ETAs, logistics, driver allocation

Result

- Google earns API revenue from every ride and delivery
- Google did NOT build Ola, Zomato, or Swiggy
- R&D cost shifted to ecosystem
- Google's ROI is higher

③ Mechanism 3 — Network Effects

Virtuous cycle:

- More developers attract more users.
- More users attract more developers.
- ⇒ **Positive feedback loop.**

⇒ **Raises firm value.**

Data Set

- 500 firms
- 55 quarters (2007 $\rightarrow$ 2020)
- 27,500 total observations (one firm at one quarter t)
- We have included macro shocks
    - Global Financial Crisis shock
    - COVID shock
    - Time trend

Firm Characteristics

- Firm size: drawn from lognormal distribution
- Industry (categorical):
  - Services: 25%
  - Manufacturing: 30%
  - Retail: 20%
  - Transport: 10%
  - Other: 20%
- Variables created:
  - `has_api` (API adoption indicator)
  - `post_api` (post-adoption indicator)
  - `time_since_adoption` (quarters since adoption)

Frame Title

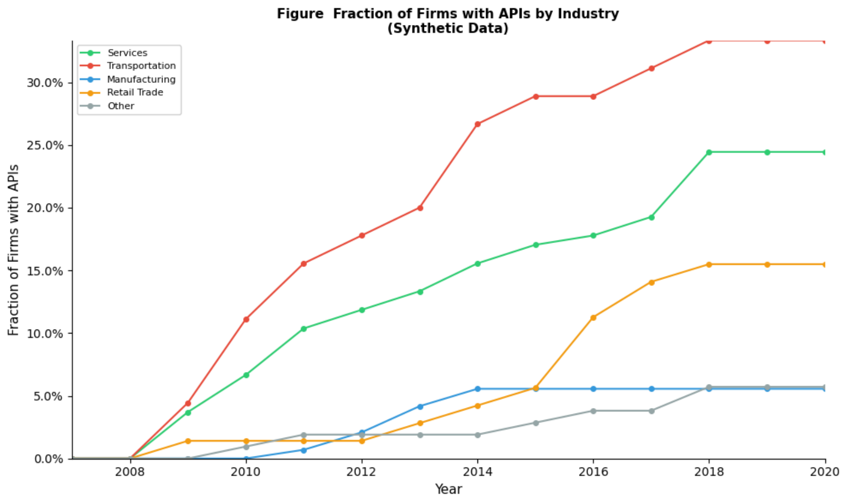
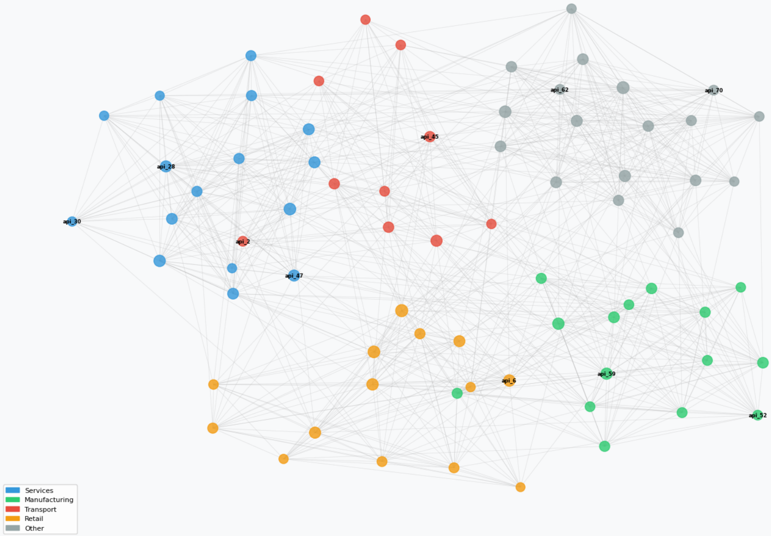


Figure Fraction of Firms with APIs by Industry
(Synthetic Data)
Services
Transportation
Manufacturing
Retail Trade
Other
Fraction of Firms with APIs
30.0%
25.0%
20.0%
15.0%
10.0%
5.0%
0.0%
2008
2010
2012
2014
2016
2018
2020
Year

Figure: API Network



Empirical Strategy

Main Hypothesis

H_0
$\beta = 0$
API adoption has no effect on firm market value.

H_1
$\beta \neq 0$ (or $\beta > 0$)
API adoption affects firm market value.

Why This Is Hard to Prove

Endogeneity Problem

- If a good firm adopts APIs, the growth may not be caused by APIs.
- The growth likely causes the adoption of API.

Key Issue
This is **endogeneity**.

Difference-in-Differences (DiD)

Core Problem

- What is the effect of API adoption on firm value?
- We never observe the same firm (at the same time):
    - adopting APIs and
    - not adopting APIs

Why do we need DiD

- API-adopting firms may already be:
    - more innovative,
    - larger,
    - better managed
- If they have higher valuations, it may have nothing to do with APIs.

How DiD Helps

- DiD differences out fixed differences.
- It does not compare levels.

Key Question
Did treated firms grow more after adoption than expected?

Difference-in-Differences Intuition

- 1st difference: removes fixed group differences
- 2nd difference: removes common time trends

Result
What remains is attributed to the treatment.

Definition
Treatment effect = Observed change − Counterfactual change

Key Assumption: Parallel Trends

Parallel Trends
Without treatment, both groups would move similarly over time.

Interpretation
The control group shows what would have happened without the treatment.

Fixed Effects Regression (DiD)

Why Fixed Effects?

- Firms difference in innovation and management.
- These affect both:
  - API adoption
  - firm value

This creates omitted variable bias, but firm fixed effects solves this.

Two-Way Fixed Effects Model

Model
$Y_{it} = \beta API_{it} + \mu_i + \tau_t + \epsilon_{it}$

- μ_i: firm fixed effects
- τ_t: time fixed effects
- Y_{it}: dependent variable
- API_{it}: API adoption indicator
- ϵ_{it}: error term

Interpreting β

Treatment Effect
β measures change in firm value after API adoption.

Example:

- If $\beta = 0.327$
- $e^{0.327} - 1 \approx 0.387$
- $\Rightarrow$ **38.7% increase**

Results

table Overall DiD Estimates

	Estimate
Overall DiD Estimate	0.1530
Percent Change	16.5%
Regression-Based DiD	0.1530
P-value	0.0004

table By Industry

Industry	DiD Estimate	Interpretation
Manufacturing	0.2035	22.6% increase
Transport	0.1917	21.1% increase
Services	0.1285	13.7% increase
Retail	0.1255	13.4% increase
Other	0.1132	12.0% increase

Event Study

Event Study: Motivation

Event study extends DiD by estimating treatment effects at multiple time periods.

Idea
Estimate treatment effects at multiple time periods.

- Not just one β
- Effects before and after treatment

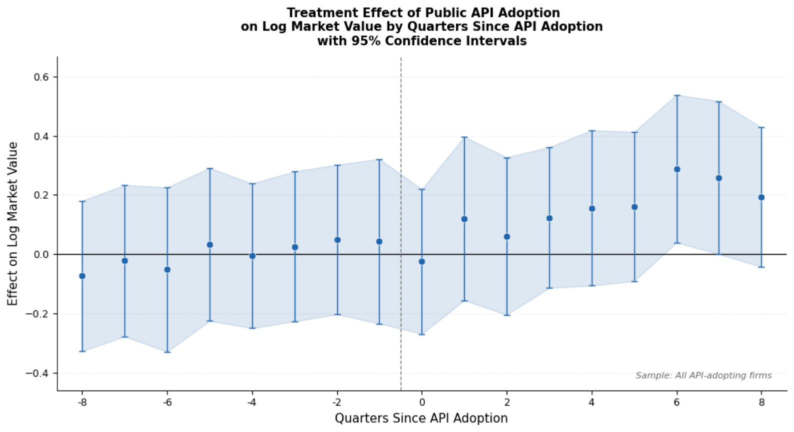
Event Study Specification

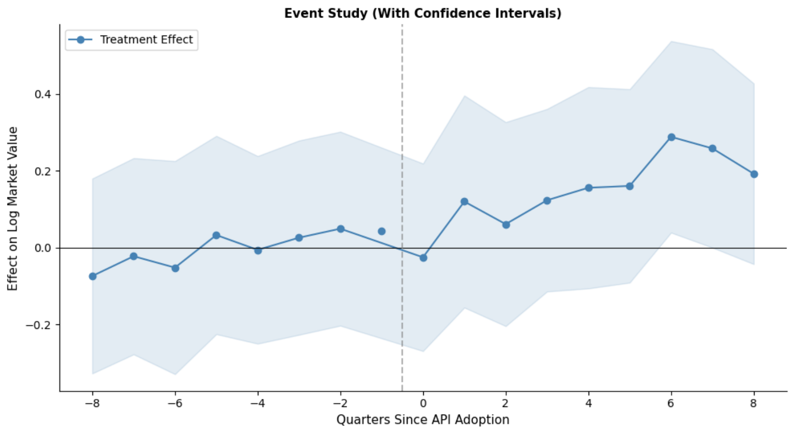
Model
$Y_{it} = \sum_{k \neq -1} \beta_k D_{it}^k + \mu_i + \tau_t + \epsilon_{it}$

- β_k: effect at time k
- $k = -1$ omitted baseline

Why Use an Event Study?

1. Test parallel trends - We look at pre-treatment coefficients, if they are near zero then there is no evidence that treated firms were already diverging.
2. Study dynamics of effects - Look at how the treatment effect changes over time (immediacy, growth, persistence).


Treatment Effect of Public API Adoption
on Log Market Value by Quarters Since API Adoption
with 95% Confidence Intervals
Effect on Log Market Value
0.6
0.4
0.2
0.0
−0.2
−0.4
Sample: All API-adopting firms
-8
-6
-4
-2
0
2
4
6
8
Quarters Since API Adoption


Event Study (With Confidence Intervals)
Treatment Effect
Effect on Log Market Value
0.4
0.2
0.0
−0.2
−8
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−4
−2
0
2
4
6
8
Quarters Since API Adoption

2SLS

2SLS Estimation

(1) First Stage

$$X_{it} = \pi_0 + \pi_1 Z_{it} + \mu_i + \tau_t + u_{it}$$

- X_{it}: endogenous variable
- Z_{it}: instrument

(2) Second Stage

$$Y_{it} = \alpha + \beta \hat{X}_{it} + \mu_i + \tau_t + \epsilon_{it}$$

- $\hat{X}_{it}$: predicted (exogenous) part
- Two stages:
  - isolate exogenous variation
  - estimate outcome effect

2SLS (Two-Stage Least Squares)

What if API adoption or network centrality is still endogenous?

Why 2SLS?

Suppose firms expecting future growth are more likely to adopt APIs, then:

$$\text{Cov}(API, \epsilon) \neq 0$$

Problem
FE estimator may be biased that is endogeneity.

Solution: Use an instrument Z that affects API adoption but does not directly affect firm value.

2SLS: Instrument Requirements

Two requirements for a valid instrument:

1. **Relevance**

$$\text{Cov}(Z, API) \neq 0$$

   The instrument must predict the endogenous regressor.

2. **Exclusion Restriction**

$$\text{Cov}(Z, \epsilon) = 0$$

   The instrument affects the outcome ONLY through API adoption.

$$\boxed{\text{Valid instrument} \rightarrow \text{Unbiased causal estimate}}$$

2SLS Results

Network Statistic	Coefficient	Std Error	F-stat
Mean Betweenness	227.24***	0.5950	0.7
Max Betweenness	136.16***	0.3565	2.0
Sum Betweenness	170.19***	0.4456	1.3
Mean Degree	276.84***	0.7249	0.5
Max Degree	187.09***	0.4899	1.1
Sum Degree	387.69***	1.0151	0.2
Mean Eff Net	-2465.52***	6.4559	0.0
Max Eff Net	1179.51***	3.0885	0.0
Sum Eff Net	-831.17***	2.1764	0.1

Positive and statistically significant coefficients ($p < 0.001$) indicate that API network centrality is associated with higher firm value.

Conclusion

API adoption increases firm value by 16.5%

- Strong statistical evidence ($p = 0.0004$)
- Network centrality positively associated with value
- Manufacturing and Transport benefit most

$\boxed{\textbf{Inverting the firm} \rightarrow \textbf{Value creation}}$

Bibliography

Van Alstyne, M., Benzell, S., & Hersh, J. (2024).
How APIs create growth by inverting the firm.
Management Science, 70(10), 7120-7141.